Rules for FDs

R1: A FD $X \rightarrow Y$ will always hold if all the values of X are unique (diff) irrespective of the values of Y.

Ex:

A	B	C	D	E
5	4	3	2	2
8	5	3	2	1
1	9	3	3	5
4	7	3	3	8

holds	doesn't hold
$A \rightarrow B$	$C \rightarrow D$
$A \rightarrow BC$	$C \rightarrow E$
$A \rightarrow CD$	
$A \rightarrow BCD$	
$A \rightarrow DE$	

In general, we can say the following FDs will always hold,

$A \rightarrow$ any combination of attributes A, B, C, D, E

Similar will be case for attributes B & E.

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1 2 3 4
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19 20 21 22 23 24 25
26 27 28 29

R2: A FD $X \rightarrow Y$ will always hold if all the values of Y are same irrespective of the values of X.

holds
$A \rightarrow C$
$AB \rightarrow C$
$ABDE \rightarrow C$
$DE \rightarrow C$
$AE \rightarrow C$

In general, we can say the following FDs will always hold,

Any combination of attributes $A, B, C, D, E \rightarrow C$

Combination of R1 & R2:

$\alpha \rightarrow \beta$ always holds if either all values of α are unique (OR) if all values of β are same (OR) both.

R3: For a FD $X \rightarrow Y$ to hold, if two tuples in the table agree on the value of attribute X, then they must also agree on the value of attribute Y.

R4: For a FD $X \rightarrow Y$, violation will occur only when for two/more same values of X, the corresponding Y values are different.

Obstacles are those frightful things you see when you take your eyes off your goal.

Practice Qus

Ex: Employee

EID	EName	ECity	EState
E001	John	Delhi	Delhi
E002	Mary	Delhi	Delhi
E003	John	Noida	U.P

EID → EName ✓
EID → ECity ✓
ECity → EState ✓
EName → EID ✗

Inference Rules / Armstrong Axioms:

I) Reflexivity:

If Y ⊂ X, then X → Y. Such FDs are trivial FDs

II) Augmentation:

If X → Y, then XZ → YZ.

III) Transitivity:

If X → Y and Y → Z, then X → Z

IV) Decomposition:

If X → YZ; then X → Y and X → Z

V) Union:

If X → Y and X → Z, then X → YZ

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vi) Pseudo transitivity Rule:

If $X \to Y$ and $YW \to Z$, then $XW \to Z$

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Q: why Armstrong axioms are sound & complete?

Q: Proof of all Armstrong axioms:

Dependency closure

A closure of a set of FDs is a set of all possible FDs that can be derived from a given set of FDs for a relation R. If F is used to denote the set of FDs for R, then the closure of a set of FDs implied by F is denoted by F+

Ex: $F = \{A \to B, B \to C, C \to D\}$

$F+ = \{A \to B, B \to C, C \to D, A \to A$ (reflexive), $A \to C$ (transitive), $A \to D$ (transitive), $A \to ABCD$ (Union) ---------- (derive rest possible FDs in this way)

Life is pleasant. Death is peaceful. It's the transition that's troublesome.

Attribute closure

Algorithm : Determining X^+, the closure of X under F.

Input : Let F be a set of FDs for rel^n R.

Steps : 1. $X^+ = X$

2. for each FD : $Y \to Z$ in F do
   If $Y \subseteq X^+$, then
   $X^+ = X^+ \cup Z$
   end if
   end for

3. Return X^+

Output : closure X^+ of X under F.

Q1 R(A, B, C, D, E, F, G) with FDs, F = { $A \to BC$, $BC \to DE$, $D \to F$, $CF \to G$ }

find closure of ABC?

$ABC^+ = (ABC) \cup (BC) = ABC$
$= (ABC) \cup (DE) = ABCDE$
$= (ABCDE) \cup (F) = ABCDEF$
$= (ABCDEF) \cup (G) = ABCDEFG$

$\Rightarrow ABC^+ = ABCDEFG$

Q1 find $A^+ = \{A\}$
$= \{A, B, C\}$ (using $A \to BC$)
$= \{A, B, C, D, E\}$ (using $BC \to DE$)

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$= \{A, B, C, D, E, F\}$ (using $D \to F$)

$= \{A, B, C, D, E, F, G\}$ (using $CF \to G$)

$\Rightarrow A^+ = \{ABCDEFG\}$

Q. find D^+ ?

$D^+ = \{DF\}$

Q. find BC^+ ?

$BC^+ = \{BCDEFG\}$

Practice Question

Q. $R(ABCDEFG)$

$F = \{AB \to CD, AF \to D, DE \to F, C \to G, F \to E, G \to A\}$

Attribute closure ?? find CF^+, BG^+ ?

$\to AF^+ = \{AFDE\}$

$\to AB^+ = \{ABCDG\}$

Q. $R(ABCGHI)$

19 SUNDAY

$F = \{A \to B, A \to C, CG \to H, CG \to I, B \to H\}$

Dependency closure, find $F^+ = ??$

First they ignore you, then they laugh at you, then they fight you, then you win

Q- Find the super keys & Candidate keys?,

R(ABCDEF)

$F = \{AB \to C, C \to D, D \to E, E \to BF, F \to A\}$

A- $(AB)^+ = \{A, B, C, D, E, F\}$ → S.K
C.K.

$A^+ = \{A\}$
$B^+ = \{B\}$

Consider,

$E \to BF$: by splitting ⇒ $E \to B$
$E \to F$

As, $E \to B$, so replace B by E, we get

$(AE)^+ = \{A, E, B, F, C, D\}$ → S.K

$A^+ = \{A\}$
$E^+ = \{E, B, F, A, C, D\}$

Consider,

$F \to A$, so replace A by F in AB,

$(FB)^+ = \{F, B, A, C, D, E\}$ → S.K

$F^+ = \{F, A\}$
$B^+ = \{B\}$

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Now consider,

$E \rightarrow BF$

$E^+ = \{A, B, C, D, E, F\}$

$D \rightarrow E$

$D^+ = \{A, B, C, D, E, F\}$

$C \rightarrow D$

$C^+ = \{A, B, C, D, E, F\}$

} C. Ks